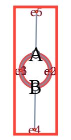
Basketball Force Layout

The force model does produce midpoint edge controls for this graph once the temporary z offsets are collapsed promptly. The old z schedule kept those hidden 3D offsets active for too long: repulsion used 3D distances, while the visible layout only drew x/y coordinates. `beta` does repel the two nodes, but it also scales edge-node, edge-edge, dangling, and center charges, so the useful tuning ratios are `k-spring` versus `beta * gamma-*`. With the dimensional scaling fixed, `length-scale` behaves as a zoom-like parameter: it changes the natural size while preserving the useful force ratios. The selected sweep point keeps the internal edge controls near their incidence midpoint and spreads them across the middle of the A-B segment.

Smoothed Curves

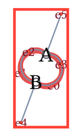
default prompt z collapse smooth edge-pos

A-B: 1.53



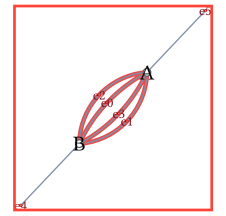
old-z-schedule delayed z collapse smooth edge-pos

A-B: 1.39



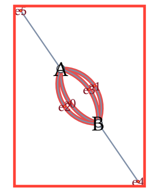
selected corrected zoom tuning smooth edge-pos

A-B: 4.61



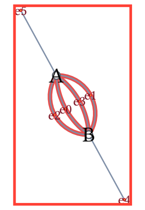
balanced-3.0 sweep target: wider nodes smooth edge-pos

A-B: 3.1



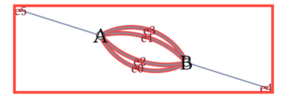
balanced-3.1 slightly longer springs smooth edge-pos

A-B: 3.17



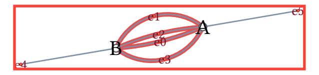
wide-3.9 longer rest length smooth edge-pos

A-B: 4.22



wide-4.1 very long rest length smooth edge-pos

A-B: 4.19

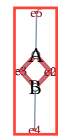


Straight Force Springs

This uses the same layout positions, but draws each paired edge as the two straight springs that the force system actually optimizes: source to edge point, then edge point to sink. That makes the old z schedule's behavior more direct: parallel edge control points repel into a crossed diamond when the hidden z coordinate is still carrying much of the separation.

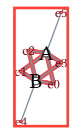
default prompt z collapse straight springs

A-B: 1.53



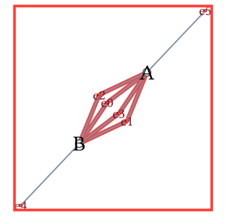
old-z-schedule delayed z collapse straight springs

A-B: 1.39



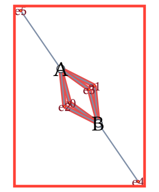
selected corrected zoom tuning straight springs

A-B: 4.61



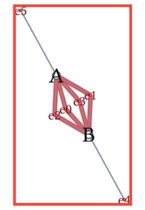
balanced-3.0 sweep target: wider nodes straight springs

A-B: 3.1



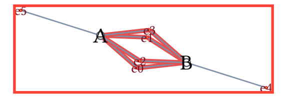
balanced-3.1 slightly longer springs straight springs

A-B: 3.17



wide-3.9 longer rest length straight springs

A-B: 4.22



wide-4.1 very long rest length straight springs

A-B: 4.19